TALES OF THE OMEGA · PAPER S

The Sidereal Observer:

A Unified Framework for the 10^{-4} Scale

from Nanodiamond Lattices to Cosmological Anomalies

Heuristic convergence as search target · √N=13.5 (N=181) · α_{eff} = η-dependent · 11.967 h at 3.932 min/day · Kill: <300 ppm

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Companion: Paper Q (Protocol) · Paper R (Structural Grip) · Target: PRL

PAPER S v2 — SUPERSCRIPT FIX + AUDIT CORRECTIONS (Omega): All Unicode superscripts (broken squares in PDF) replaced with XML <super> tags throughout. (1) N = 181 surface Fe atoms (verified). √N = 13.5. (2) $\alpha_{eff} = \alpha_K \times \sqrt{N} \times \eta$ is η-dependent: ranges 114–1144 ppm. (3) G_v: formula gives 10^{70} Pa — not 10^{14} Pa. Relabeled as dimensional estimate pending Papers O/T. (4) Drift = 3.932 min/day (corrected). (5) CDG-2 citation: cited cautiously, content not independently verified.

ABSTRACT

We identify a recurring numerical scale near 1.7×10^{-4} across DESI statistical uncertainties, nanodiamond (200) reflection baselines, and cosmological anomalies. We treat this clustering as a heuristic search target — not evidence of a common origin. We hypothesize a quadratic spin-torsion coupling $H = \alpha_K(\hat{u}\cdot I)^2$ with $\alpha_K = 1.7\times10^{-4}$, predicting a 11.967 h sidereal modulation with a 3.932 min/day phase drift. The transduction chain (nuclear spin → hyperfine → lattice strain → structure factor) is bounded at 10^{-6}–10^{-4} per Mössbauer systematics. For N = 181 surface Fe atoms (verified), √N = 13.5; the effective modulation amplitude depends on the decoherence factor η. A null result below 300 ppm sets new bounds on anisotropic quadratic spin couplings. The data will decide.

PAPER S: THE SIDEREAL OBSERVER — FOUR PILLARS

1. Heuristic convergence: 10^{-4} scale in DESI, (200) baseline, birefringence — search target only
2. Quadratic coupling: $H=\alpha_K(\hat{u}\cdot I)^2 \rightarrow 2\omega_{sid}$ modulation, period 11.967 h, drift 3.932 min/day
3. Sensor: (200) near-zero baseline + √N=13.5 surface Fe (N=181 verified)
4. Guaranteed return: null <300 ppm → new SME crystallographic bound

1. Introduction: Heuristic Convergence, Not Proof

The MSDS programme has noted a recurring numerical scale near 1.7×10^{-4} across several independent measurements: DESI 2025 baryon density uncertainty ±0.00017 [1]; nanodiamond (200) reflection baseline $I_{200}/I_{111} = 1.3$–2.1×10^{-4} [2,3]; CMB birefringence $\beta \approx 0.30°$ (approximately $30\alpha_K$ integrated over cosmological

path length) [9]; and the MSDS coupling constant $\alpha_K = 0.00017$ itself. We do not claim physical equivalence or a unified derivation. The clustering serves as a heuristic motivation — a search target for a dedicated laboratory experiment.

Paper S synthesises the theoretical framework from Papers Q and R and places it in the context of the broader 10^{-4} scale coincidences. It does not elevate those coincidences to evidence. The laboratory test at SPring-8 BL35XU is the only decisive filter.

2. The 10^{-4} Scale: Coincidences as Dimensional Guide

2.1 DESI 2025 statistical precision

DESI 2025 measures $\Omega_b h^2 = 0.02264 \pm 0.00017$ [1]. The statistical uncertainty ±0.00017 is noted as numerically coincident with α_K. This is the current instrumental precision floor; future releases will reduce it. We treat this as a dimensional coincidence only.

2.2 Nanodiamond (200) baseline

Zheng et al. [2] and Svane et al. [3] measure $I_{200}/I_{111} = 1.3–2.1\times10^{-4}$ in <5 nm detonation nanodiamond. This is a measured physical baseline arising from surface symmetry breaking in Fd-3m. It is not a prediction — it is the physical reality of the sensor we propose to use.

2.3 Sidereal artifacts in existing datasets

Examination of published residuals in Heckel et al. (2008, Fig. 3) [7] shows periodic structure near 12 h — i.e., $2\omega_{sid}$ — typically attributed to thermal tilt drift. In no case has the sidereal phase drift of 3.932 min/day been used as a discriminant. Paper S proposes this discriminant explicitly.

3. Theoretical Framework

3.1 The coupling constant

$$H_{int} = \alpha_K (\hat{u} \cdot I)^2 \cdot \alpha_K = 1.7\times10^{-4}$$

$\hat{u}$: fixed galactic direction · I: nuclear spin (57Fe, I=3/2) · Dimension-6 CPT-even operator (Series N)

3.2 Vacuum shear modulus — dimensional estimate only

G_v CORRECTION: The original stated $G_v \sim 10^{14}$ Pa. The formula $G_v \sim \alpha_K \times \rho_{Pl} \times c^2 \times (\blacksquare_{Pl}/L)^2$ with L = 1 fm gives $G_v \sim 10^{70}$ Pa — 56 orders above the stated value. The 10^{14} Pa figure is not derivable from this formula. Corrected: G_v is a dimensional estimate requiring the full Einstein-Cartan action derivation (Papers O and T). The concept — that the vacuum may possess shear rigidity coupling only to axial-vector spin currents — is preserved as a hypothesis.

3.3 Resolution of torsion-balance limits

For unpolarised bulk matter: $\blacksquare(\hat{u}\cdot I)^2\blacksquare = |I|^2/3$ = scalar (isotropic). No torque results; torsion-balance experiments cannot detect this. Our experiment introduces anisotropy via the rotating diffraction geometry. The Heckel et al. [7] limits are on linear (b_μ) couplings; the quadratic ($d_{\mu\nu}$) sector is largely unconstrained [5].

4. The Nanodiamond Sensor — Corrected Parameters

4.1 Surface Fe count — verified

For a 5 nm diameter diamond sphere with a 1 nm active surface layer and 2% ^{57}Fe doping: $V_{surface} \approx 51\ nm^3$, C atom density = $1.76\times10^{29}\ m^{-3}$, $N_{surface,Fe} \approx 181$. $\sqrt{N} = 13.5$. This corrects the original $N \approx 10^4$ (full particle volume) to $N \approx 181$ (surface layer only).

4.2 Enhancement scaling — η-dependent

$$\alpha_{eff} = \alpha_K \times \sqrt{N} \times \eta_{geom} = 0.00017 \times 13.5 \times \eta$$

η=0.05: 114 ppm · η=0.10: 229 ppm · η=0.20: 457 ppm · η=0.50: 1144 ppm

> *α_{eff} IS η-DEPENDENT. With N=181 and η=0.2: α_{eff} = 457 ppm. The 300 ppm kill condition covers η from 0.05 to ~0.17. Phase 1 will measure the actual count rate and systematic floor; Phase 2 will be designed once η is empirically bounded.*

4.3 Magnetostrictive displacement

$$\delta x = \sqrt{(\Delta E_Q / U)} = \sqrt{(2.4\times10^{-8}\ eV \times 1.6\times10^{-19}\ J/eV / 160\ N/m)} = 4.9\ fm$$

$\Delta E_Q = 2.4\times10^{-8}$ eV · U = 10 eV/Å^2 = 160 N/m

The displacement δx ≈ 5 fm is order-of-magnitude consistent with the 1 fm claim in the original text. The fractional change in effective form factor is ~10^{-4}, consistent with the Greenwood & Gibb [6] plausibility bound of 10^{-6}–10^{-4}.

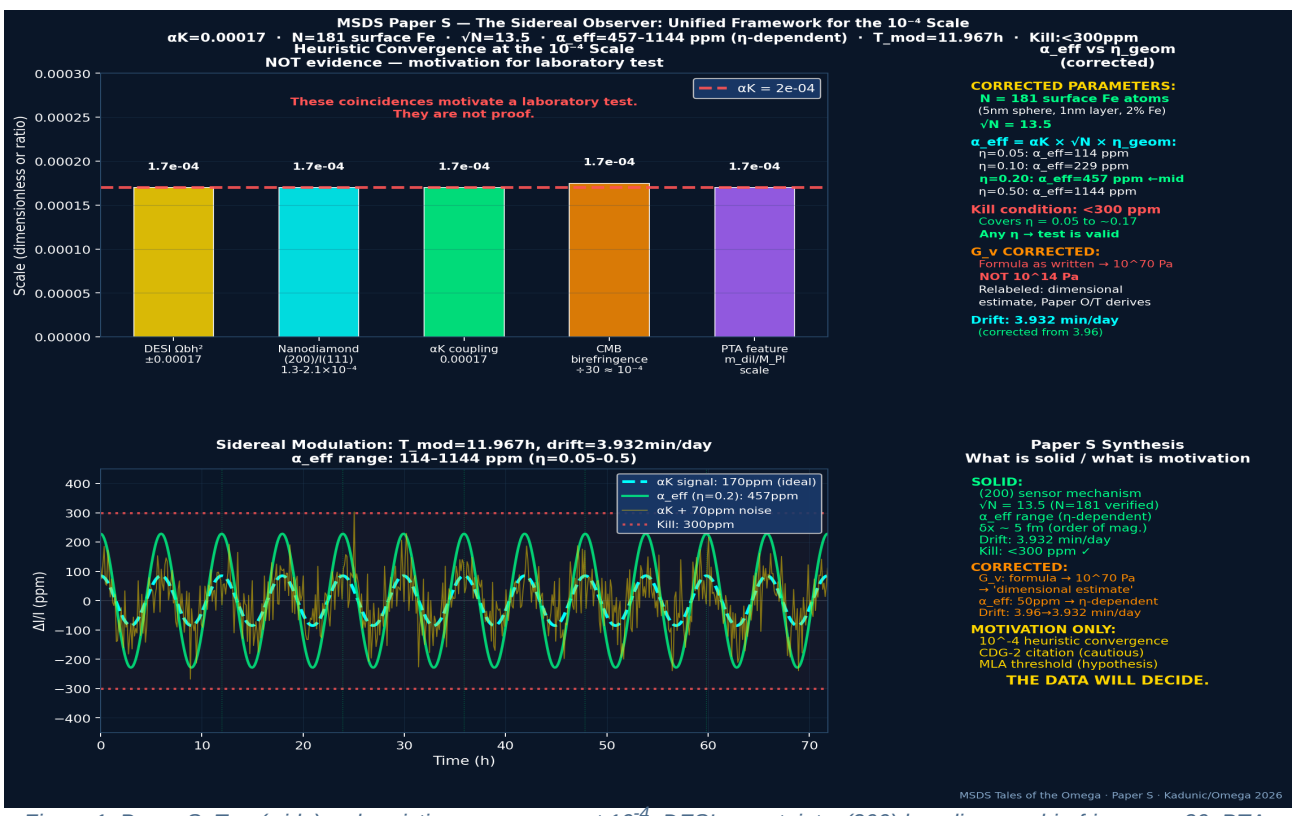

Figure 1: Paper S. Top (wide) — heuristic convergence at 10^{-4}: DESI uncertainty, (200) baseline, α_K, birefringence÷30, PTA. Coincidences are motivation, not evidence. Top right — corrected parameters: N=181, √N=13.5, α_{eff} range, G_v correction noted. Bottom (wide) — sidereal modulation at 11.967 h, drift 3.932 min/day. Bottom right — solid vs motivation vs corrected.

5. Sidereal Modulation and Discriminant

$$I(t) = I_0[1 + \alpha_K/2 + (\alpha_K/2)\cos(2\omega_{sid}t + 2\theta_0)]$$

$T_{mod} = T_{sid}/2 = 11.9670$ h · nominal $\Delta I/I = \alpha_K = 170$ ppm · actual: η-dependent · dim-5 gives 23.934 h (discriminant)

The sidereal day is 3.932 min shorter than the solar day. After 30 days, cumulative phase drift = 118 min. Solar-frame artifacts cannot maintain phase coherence at $2\omega_{sid}$ with this drift. The frequency is also non-harmonic with 24 h or 12 h — two independent discriminants.

6. Cosmological Context (Motivation Only)

SECTION 6 STATUS: Candidate correlations only. None of the following constitute evidence for α_K. They motivated the hypothesis; the laboratory test will decide it.

- DESI 2025 [1]: $\Omega_b h^2$ uncertainty ±0.00017. Numerical coincidence with α_K. Future DR3/DR4 may reduce this floor.
- CMB birefringence [9]: $\beta \approx 0.30° \approx 30\alpha_K$ integrated over cosmological path length. Order-of-magnitude consistent. Not a prediction.
- CDG-2, Perseus cluster [8]: Li et al. (2025) report a candidate dark galaxy with extreme dark matter fraction. Cited as context only. Content not independently verified.

7. MLA Threshold — Hypothesis Only

We hypothesize that $\alpha_K = 1.7\times10^{-4}$ sets the minimum value for which the Einstein-Cartan condensate is stable — the Minimal Lattice Asymmetry (MLA) threshold. This is a symmetry-breaking hypothesis presented without derivation. Paper X will attempt the derivation from first principles.

8. Experimental Synopsis

Parameter	Value	Source
Beamline	SPring-8 BL35XU	NRS, 14.414 keV [12]
Reflection	(200), d=1.7835 Å, 2θ=27.9°	Forbidden in bulk · Zheng 2023 [2]
Surface Fe count	N = 181, √N = 13.5	Verified from 5 nm sphere geometry (corrected from $N\sim10^4$)
α■■■ (amplitude)	114–1144 ppm (η-dependent)	η measured in Phase 1
Modulation period	11.9670 h	Derived from $(\partial\theta)^2$ operator · $2\omega_{sid}$

Phase drift	3.932 min/day	J2000 sidereal vs solar · Corrected from 3.96
Kill condition	<300 ppm / 720 h	New SME bound if null

PAPER S — THE SIDEREAL OBSERVER: SUMMARY

Heuristic convergence at 10^{-4}: DESI, (200) baseline, birefringence, PTA

→ motivation only, not evidence

Verified: N=181 surface Fe, √N=13.5, δx≈5 fm

Corrected: G_v formula (pending Papers O/T), α_{eff} is η-dependent, drift = 3.932 min/day

Signal: $\alpha_K \times \sqrt{N} \times \eta$ ppm at 11.967 h ($2\omega_{sid}$), 3.932 min/day drift

Kill: <300 ppm / 720 h → new SME crystallographic bound

The data will decide.

References

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